William Qian

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Education

Harvard University 2023–2028 (expected)
Ph.D. candidate in Biophysics; advisor: Prof. Cengiz Pehlevan
Graduate Fellow, Kempner Institute for Natural and Artificial Intelligence
Research interests: recurrent neural networks, computational neuroscience, neural network interpretability

University of Pennsylvania 2018–2022
M.S. and B.S. in Physics; B.S. in Computer Science

Publications

- **Efficiency and robustness partition the solution space for memory in recurrent networks**
  In Submission
  W. Qian, C. Pehlevan
- **Discovering alternative solutions beyond the simplicity bias in recurrent neural networks**
  ICLR 2026
  W. Qian, C. Pehlevan
- **Deep RL Needs Deep Behavior Analysis: Exploring Implicit Planning by Model-Free Agents in Open-Ended Environments**
  NeurIPS 2025
  R. Simmons-Edler, R. P. Badman, F. B. Berg, R. Chua, J. J. Vastola, J. Lunger, **W. Qian**, K. Rajan
- **Neurocomputational underpinnings of suboptimal beliefs in reinforcement learning agents**
  Cognitive Computational Neuroscience (CCN), 2025
  M. G. Kumar, A. Manoogian, **W. Qian**, C. Pehlevan, S. A. Rhoads
- **Partial observation can induce mechanistic mismatches in data-constrained models of neural dynamics**
  NeurIPS 2024
  W. Qian, J. Zavatone-Veth, B. S. Ruben, C. Pehlevan
- **Optimizing the human learnability of abstract network representations**
  Proceedings of the National Academy of Sciences (PNAS), 2022
  W. Qian, C. W. Lynn, A. A. Klishin, J. Stiso, N. H. Christianson, D. S. Bassett
- **Path-dependent dynamics induced by rewiring networks of inertial oscillators**
  Physical Review E, 2022
  W. Qian, L. Papadopoulos, Z. Lu, K. A. Kroma-Wiley, F. Pasqualetti, D. S. Bassett

Work Experience

PathAI (acquired by Roche) — *Machine Learning Engineer*, Boston, MA Jun. 2022–Aug. 2023
Designed and trained end-to-end models for detecting oncogenic mutations from gigapixel histology slide images.

PathAI — *Machine Learning Intern*, Boston, MA Jun.–Sept. 2021
Conducted machine-learning research on discouraging shortcut learning in the low-data regime.

Invited Talks & Presentations

- **Mechanistic biases in data-constrained models of neural dynamics**
  Janelia Workshop on Grounding Cognition in Mechanistic Insight, 2025
  Houston Neurotheory Seminar, 2025
- **Optimizing the human learnability of abstract network representations**
  DCIST Collaborative Research Alliance, 2022
- **When it is more powerful to infer than to see**
  American Physical Society March Meeting, 2021

Teaching & Service

Teaching Fellow, *Harvard University*
Introduction to Applied Mathematics — Spring 2025

Teaching Assistant, *University of Pennsylvania*
Introduction to Machine Learning — Fall 2021, Spring 2022
Mathematical Foundations of Computer Science — Fall 2019, Spring 2020, Fall 2020
Ad hoc reviewer: NeurIPS, ICLR, ICML, Nature Communications

Skills

Programming: Python, Java; familiarity with C++, MATLAB
ML / Scientific Computing: PyTorch, JAX, NumPy, pandas, scikit-learn, Mathematica
Tools: Git, Linux, Slurm

Awards & Honors

Kempner Graduate Fellowship — 2023–Present
Roy and Diana Vagelos Molecular Life Scholar — 2022
Roy and Diana Vagelos Research Stipend — 2020–2021
Dean's List — 2018–2022
USA Astronomy and Astrophysics Olympiad, Team Selection Camp — 2017